Carl Abou Saada Nujaim

1st year PhD student in Human-Computer Interaction at Université Paris-Saclay with a background in Computer Science and Animation.

Supervised by Wendy Mackay & Michel Beaudouin-Lafon

carl@lisn.fr
Antony, France
carlnujaim.com
Carl Abou Saada Nujaim
@carlnujaim

Research interests

HCI Creativity-Support Tools
Instrumental Interaction
Interaction Substrates
Frontend Development
Designer-Developer Workflow
Visual Programming
Creative collaboration

Languages

French Arabic English

Programming Languages

JavaScript/TypeScript C#
Java Dart C/C++ Python
HTML/CSS PHP SQL Bash

Publications

"Figma is a big black box:" The mismatch between the interface and its underlying structure (poster)

ACM CHI · Apr. 2026 · Link

Research experience

ex)situ, LISN

Orsay, France · Oct. 2025 → Oct. 2028 · Website

Pursuing a PhD to explore and enhance the creation process of interactive systems, from both the designer and the developer sides, and strengthen the collaboration between them. My current project is a collaborative frontend development toolkit grounded in HCI theory.

ex)situ, LISN

Orsay, France · May → Aug. 2024 & Mar. → Sept. 2025 · Website

Completed two research internships in the ex)situ research team where I participated in the design and development of a new version of *VideoClipper (Mackay et al., 2024)*, utilizing interaction substrates (Mackay & Beaudouin-Lafon, 2025) to extend the tool's power of expression. I conducted a user study and interviews.

Cre8mania Digital Agency

Beirut, Lebanon · Feb. → May 2022 · Website

Research internship where I studied different AR modalities and their characteristics, developed image tracking and markerless AR prototypes with various web technologies, and tested them and compared their efficiency and power of expression.

Projects

TurboDOMBuilder

Feb. 2024 → Present · Link

An in-development frontend TypeScript library that incorporates HCI concepts such as instruments and substrates, and introduces an MVC-inspired architecture that provides varying levels of structure, while still maintaining flexibilitiy.

eave - The new shopping experience

Feb → May. 2023 · Link

Created a 3D virtual and collaborative shopping environment in Unity. Designed and implemented the environmental (3D) and information-related (2D) interactions.

Child bullying visualization

Mar. → Apr. 2024 · Link

Designed and developed with a partner an interactive visualization that depicts the percentage of bullied kids in Argentina in 2018 based on different physical and social characteristics.

Invisible haunted mansion

Sept. 2023 → Oct. 2023 · Link

An immersive ghost-hunting game created in a group project and designed to be non-visual and leverage other senses. The player catches ghosts in their physical room using their smartphone/ wearable device with auditory and haptic feedback.

Upool carpooling system

Dec. 2022 · Link

Designed and user-tested a transportation solution for university students in Lebanon based on user interviews.

*For a more comprehensive list, take a look at the projects on my **Linkedin** profile.*

Education

Université Paris-Saclay Orsay, France Sept. 2023 → Sept. 2025
Master's degree in Human-Computer Interaction — Average: 16.7/20, ranked 1st.

Lebanese American University Beirut, Lebanon Sept. 2020 → June 2023
BS in Computer Science with a minor in Animation — CGPA: 3.95/4.

Collège Notre-Dame de Jamhour Beirut, Lebanon Sept. 2007 → June 2020
Grade 12 average: 17.3/20, ranked 3rd.

Awards

IDEX Scholarship Sept. 2023 Université Paris-Saclay

Murex Award June 2023 Murex
Most Collaborative Development Project Award for my Capstone Project "eave".

Honor Scholarship Sept. 2020 Lebanese American University

Test scores

TOEFL 112/120 Dec. 2023

Baccalauréat Français 17.73/20 June 2020

SAT 1510/1600 Dec. 2018

Design tools

Figma Adobe Creative Suite
Maya Revit 3DS Max
Blender Toon Boom Harmony

Programming tools

Unity Flutter React OpenGL
THREE.js Oracle MongoDB

Soft skills

Creativity Problem Solving
Attention to Details
Organization Teamwork
Time Management
Self-learning

Hard skills

User Interviews
UX Design Methods
Web & Mobile Development
Game Development
Software Documentation
Software Design
3D Modeling & Animation
Motion Design Cell Animation